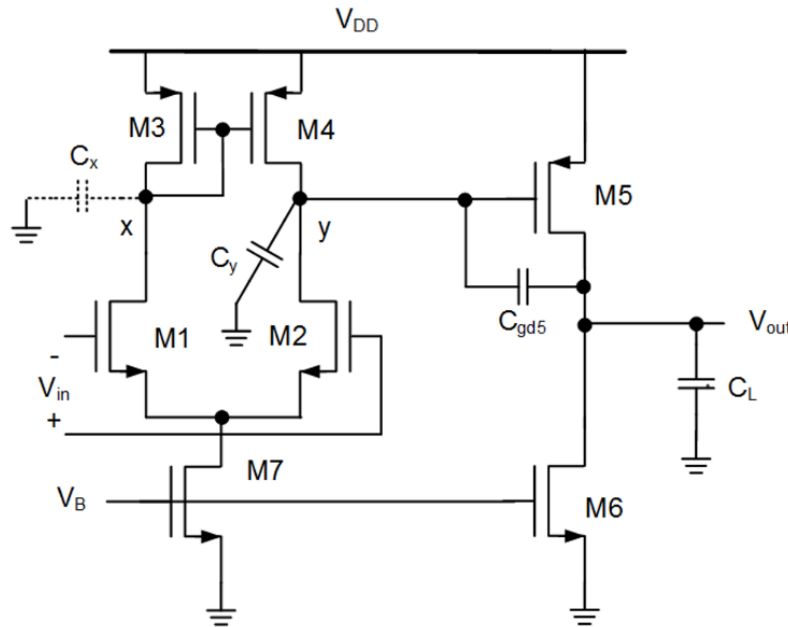


Q1. Design a two-stage op-amp shown below, where $V_{DD} = 3V$, total power consumption=6mW, maximum output swing=2.6V, channel length= $0.5\mu m$, $V_{THN}=0.7V$, $V_{THP}=-0.8V$, $\lambda_n=0.1V^{-1}$, $\lambda_p=0.2V^{-1}$, $k'_n=134\mu A/V^2$, $k'_p=38.3\mu A/V^2$, Ignore channel modulation effect for DC calculation and body effect.

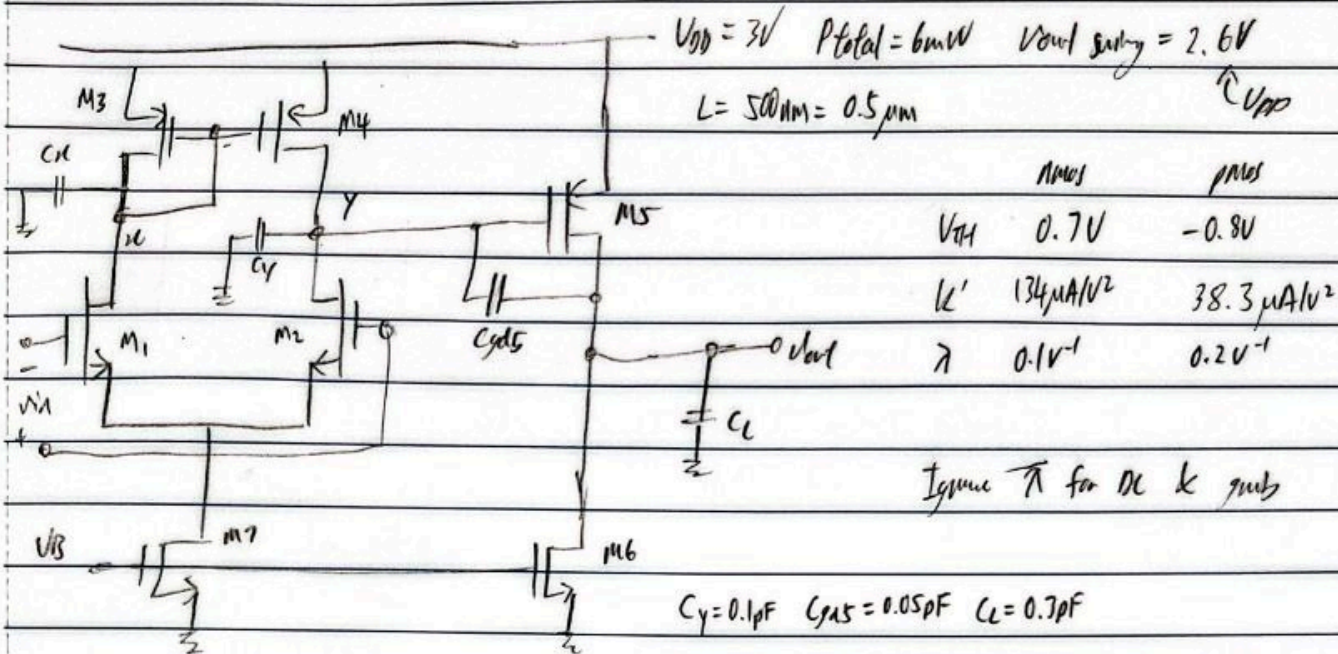
- (1) Assume 1mA current is allocated to the second stage and equal overdrive voltage for M5 and M6, determine the aspect ratios (W/L) of these two transistors;
- (2) Calculate small-signal gain of the second stage;
- (3) Assuming the remaining 1mA flows through M7, find the aspect ratios (W/L) of M3 and M4 so that $V_{GS3}=V_{GS5}$ to minimize the systematic offset.
- (4) Calculate the aspect ratios (W/L) of M1 and M2 such that the overall voltage gain of the op-amp is 1000.
- (5) Find the pole and the zero frequencies, and obtain the transfer function by considering only C_y , C_{gd5} , and C_L .



* $C_y = 0.1pF$, $C_{gd5} = 0.05pF$, $C_L = 0.3pF$

Fig. 1

EE 3408C twl4



1) $I_{D5} = I_{D6} = 1mA$ $V_{GS5} = V_{GS6}$

min swing = 2.6Vpp $\Rightarrow V_{GS5} = V_{GS6} = \frac{3-2.6}{2} = 0.2V$

$I_D = \frac{1}{2} (\frac{W}{L}) K' V_{GS}^2$ [ignore λ at DC]

$(\frac{W}{L}) = \frac{2I_D}{K' V_{GS}^2}$

$(\frac{W}{L})_5 = \frac{2 \times 1m}{38.3\mu A \times 0.2^2}$ $(\frac{W}{L})_6 = \frac{2 \times 1m}{134\mu A \times 0.2^2}$

≈ 1305

≈ 373

4) $A_{v1} = 1000 \Rightarrow A_{v1} = \frac{1000}{(100/3)} = -30 V/V$

$A_{v1} = -g_{m2} (r_{o4} || r_{o2})$

$= -\sqrt{2I_{D2} (\frac{W}{L})_2 K'} \times \frac{1}{\lambda_{nI_{D2}} + \lambda_{pI_{D4}}}$

$-30 = -\sqrt{2 \times 0.5mA \times (\frac{W}{L})_2 \times 134\mu A} \times \frac{1}{(0.1+0.2) \times 0.5mA}$

$4.5 \times 10^{-3} = \sqrt{1.34 \times 10^{-7} (\frac{W}{L})_2}$

$(\frac{W}{L})_2 = \frac{10125}{67} \approx 151 \#$ matched $\Rightarrow (\frac{W}{L})_1 \approx 151 \#$

2) $A_{v2} = -g_{m5} (r_{o5} || r_{o6})$ $r_o = \frac{1}{\lambda I_D}$

$= -\frac{2I_{D5}}{V_{GS5}} \times \frac{r_{o5} r_{o6}}{r_{o5} + r_{o6}}$

$= -\frac{2I_{D5}}{V_{GS5}} \times \frac{1}{\lambda_{nI_{D5}} + \lambda_{pI_{D6}}}$

$= -\frac{2}{V_{GS5}} \times \frac{1}{\lambda_{n1} + \lambda_{p2}}$

$= -\frac{2}{0.2} \times \frac{1}{0.1+0.2}$

$= -100/3$

$\approx -33.3 V/V$

$\approx 30.5 dB$

5) since C_{gd5} is small (ie no compensation added)

$\Rightarrow C_1 = C_Y + C_{gd5} (1 - A_{v2})$ $C_2 = C_L + C_{gd5} (1 - \frac{1}{A_{v2}})$

$= 0.1 + 0.05 (1 - \frac{100}{3})$

$= 109/60 \approx 1.82 pF$

$= 0.3 + 0.05 (1 - \frac{3}{100})$

$= 0.3515 pF$

$\omega_{p1} = \frac{1}{C_1 (r_{o4} || r_{o2})}$ $\omega_{p2} = \frac{1}{C_2 (r_{o5} || r_{o6})}$ $\omega_{z1} = \frac{g_{m5}}{C_{gd5}}$

$= \frac{60 \times 10^{-3}}{1M} \times (0.1+0.2) \times 0.5 \times 10^{-3}$ $= \frac{1}{0.3515 pF} \times (0.1+0.2) \times 1 \times 10^{-3}$ $= \frac{2 \times 1mA}{0.2V} \times \frac{1}{0.05 pF}$

$\approx 8.26 \times 10^7 rad/s$ $\approx 8.53 \times 10^8 rad/s$ $\approx 2 \times 10^{11} rad/s$

$f_{p1} \approx 1.31 \times 10^7 Hz$ $f_{p2} \approx 1.36 \times 10^8 Hz$ $f_{z1} \approx 3.18 \times 10^{10} Hz$

3) $I_{D7} = 1mA \Rightarrow I_{D3} = I_{D4} = I_{D1} = I_{D2} = 0.5mA$

to have $V_{GS3} = V_{GS5} \Rightarrow (\frac{W}{L})_3 = \frac{1}{2} (\frac{W}{L})_5$

≈ 652.5

matched $\Rightarrow (\frac{W}{L})_4 = (\frac{W}{L})_3 \approx 652.5 \#$

$H(s) = \frac{v_{out}(s)}{v_{in}(s)} = 1000 \frac{(1 - \frac{s}{2 \times 10^{11}})}{(1 + \frac{s}{8.26 \times 10^7})(1 + \frac{s}{8.53 \times 10^8})} \#$

- can estimate from bode the unit gain bandwidth is at 13 GHz

Q2. For the feedback amplifier shown in Fig. 2 below, assume that the feedback coefficient, β , is independent of frequency and the open-loop transfer function of the operational amplifier is

$$A(s) = \frac{A_{DC}}{1 + s/\omega_{3dB}}$$

- (1) Find the unity-gain bandwidth (bandwidth at gain = 1) of the op-amp
- (2) Derive the closed-loop transfer function, $A_{cl}(s) = V_{out}(s)/V_{in}(s)$.
- (3) Find the 3dB bandwidth of the feedback amplifier and comment on the result.
- (4) Show that the gain-bandwidth product of the closed-loop amplifier is a constant and equals to its unity-gain bandwidth of the op-amp.

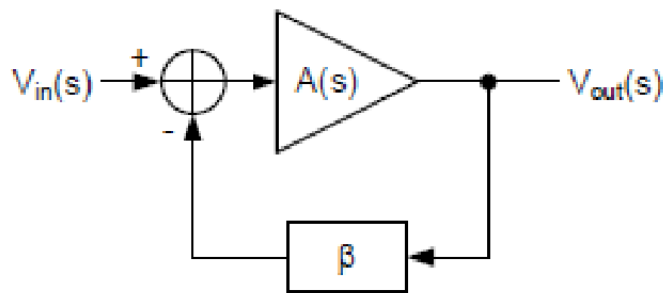
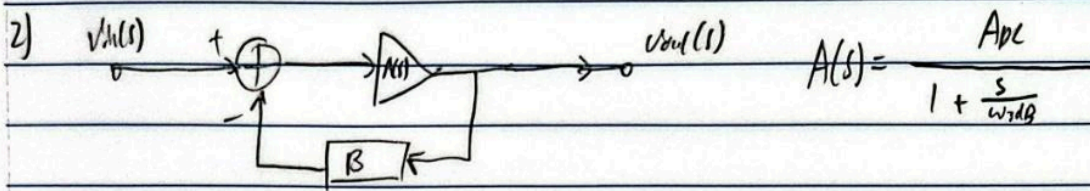
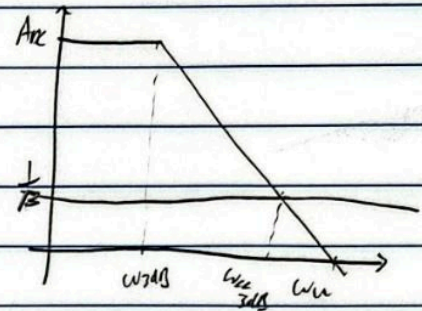


Fig. 2



i) $|A(j\omega)| = \frac{A_{DC}}{\sqrt{1 + \left(\frac{\omega}{\omega_{3dB}}\right)^2}} = 1$
 $\downarrow$ Assume $A_{DC} \gg 1 \Rightarrow \omega_{u} \gg \omega_{3dB}$
 $\frac{A_{DC}}{\omega_u / \omega_{3dB}} = 1$
 $\omega_u = A_{DC} \omega_{3dB} \#$



ii) $V_{out}(s) = A(s) [V_{in}(s) - V_{out}(s)B]$

$$A_{CL}(s) = \frac{V_{out}(s)}{V_{in}(s)} = \frac{A(s)}{1 + A(s)B}$$

$$= \frac{A_{DC}}{1 + \frac{s}{\omega_{3dB}} + A_{DC}B}$$

$$= \frac{\frac{A_{DC}}{1 + A_{DC}B}}{1 + \frac{s}{\omega_{3dB}(1 + A_{DC}B)}} \#$$

iii) compare to $\frac{A_{DC}}{1 + \frac{s}{\omega_{3dB}}}$

$$\Rightarrow A_{CL,DC} = \frac{A_{DC}}{1 + A_{DC}B}$$

$$\Rightarrow \omega_{CL,3dB} = \omega_{3dB}(1 + A_{DC}B) \#$$

3dB BW increased by $(1 + A_{DC}B)$ at the expense of
 DC gain decreased by $(1 + A_{DC}B)$

iv) $GBW = A_{CL,DC} \times \omega_{CL,3dB}$

$$= \frac{A_{DC}}{1 + A_{DC}B} \times \omega_{3dB}(1 + A_{DC}B)$$

$$= A_{DC} \omega_{3dB} \# \text{ (same as GBW of } A(s) \text{) \#}$$

GBW is a constant since both A_{DC} and ω_{3dB} are constant